

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 31)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

Answer all questions in the spaces provided.

0 1

Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 1

. 1

Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

| Mechanism                                                                      | Type of feedback |  |
|--------------------------------------------------------------------------------|------------------|--|
| Higher temperatures increase cloud cover, raising albedo                       |                  |  |
| Melting permafrost releases methane                                            |                  |  |
| Higher CO <sub>2</sub> levels increase photosynthesis and carbon sequestration |                  |  |
| Melting ice reduces albedo                                                     |                  |  |

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0 1

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Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

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Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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0 1 . 4

Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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0 2

Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

| Number of nights             | 1  | 2  | 4  | 6  | 8  | 10 | 12 | 14 |
|------------------------------|----|----|----|----|----|----|----|----|
| Running mean catch per night | 31 | 18 | 24 | 21 | 22 | 22 | 21 | 22 |

0 2 . 1

Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

0 2 . 2

Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

0 2 . 3

Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

0 2 . 4

Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 3 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

**Timber** .....

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**Fuel** .....

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**Food** .....

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**Fibres** .....

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**Medicines** .....

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**03 . 2**

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

**[3 marks]**

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**03 . 3**

Explain how forests regulate the hydrological cycle and local climate.

**[2 marks]**

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04 . 1

Describe how agriculture alters the hydrological cycle.

[3 marks]

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04 . 2

Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

**Seasonal / local food** .....

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**Organic food** .....

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04 . 3

Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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0 5 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar

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B Acoustic / satellite tag

☐

C Pitfall trap

☐

D Geolocator tag recovered by recapture

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0 5 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 6

The Russell formula describes changes in a fish stock:  $S_2 = S_1 + (A + G) - (C + M)$ , where  $S_1$  and  $S_2$  are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

| Term  | Biomass / thousand tonnes |
|-------|---------------------------|
| $S_1$ | 600                       |
| A     | 90                        |
| G     | 150                       |
| C     | 220                       |
| M     | 80                        |

0 6 . 1

Calculate  $S_2$  and state whether the stock is being fished sustainably.

[2 marks]

**Answer**  thousand tonnes

**06 . 2** Calculate the maximum catch that would have kept the stock constant. **[1 mark]**

**Answer**  thousand tonnes

**06 . 3** Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult. **[3 marks]**

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**06 . 4** Describe how each of the following measures obtained from commercial catches can indicate overfishing. **[4 marks]**

**Catch per unit fishing effort** .....

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**Mean fish size and mean age** .....

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07 . 1

Define each of the following ecological terms and give an example.

[8 marks]

**Species** .....

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**Ecological niche** .....

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**Community** .....

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**Biome** .....

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**0 7 . 2**

Explain the difference between an ecosystem and a biome.

**[2 marks]**

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**10****0 8**

Table 1 shows some features of Earth, Mars and Venus.

**Table 1**

| <b>Feature</b>                         | <b>Venus</b> | <b>Earth</b> | <b>Mars</b> |
|----------------------------------------|--------------|--------------|-------------|
| Mean distance from Sun / million km    | 108          | 150          | 228         |
| Mass relative to Earth                 | 0.82         | 1.0          | 0.11        |
| Atmospheric pressure relative to Earth | 92           | 1.0          | 0.006       |
| Mean surface temperature / °C          | 460          | 15           | -60         |
| Magnetic field                         | Very weak    | Strong       | Very weak   |
| Rotation period / hours                | 5 800        | 24           | 25          |

**0 8 . 1**

Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

**[3 marks]**

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**0 8 . 2**

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

**[3 marks]**

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08 . 3

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

08 . 4

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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09 . 1

Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

09 . 2

Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

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09 . 3

Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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10 . 1

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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Extra space

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**1 0 . 2**

State **one** reason why rainforest conservation is important to countries far from the tropics.

**[1 mark]**

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**10**

**Write an essay on one of the following topics.**

$$\boxed{1}\boxed{1}.\boxed{1}$$

Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

**[25 marks]**

**OR**

1 1 . 2

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.



Do not write outside the box

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**END OF QUESTIONS**